LAB 8

Implementation of Huffman Coding

C++ Code

```
#include <iostream>
#include <queue>
#include <vector>
#include <string>
using namespace std;

struct Node {
    char ch;
    int freq;
    Node *left, *right;

    Node(char c, int f) : ch(c), freq(f), left(nullptr), right(nullptr) {}
};

struct Compare {
    bool operator()(Node* a, Node* b) {
        return a->freq > b->freq;
    }
};

void printCodes(Node* root, string code) {
    if (!root) return;

    if (!root->left && !root->right) {
        cout << root->ch << ": " << (code.empty() ? "0" : code) << '\n';
        return;
    }

    printCodes(root->left, code + "0");
    printCodes(root->right, code + "1");
}

int main() {
    int n;
    cout << "Enter number of characters: ";
    cin >> n;

    vector<Node*> nodes;
    priority_queue<Node*, vector<Node*>, Compare> pq;

    cout << "Enter character and frequency:\n";
    for (int i = 0; i < n; ++i) {
        char ch;
        int freq;
        cin >> ch >> freq;
        pq.push(new Node(ch, freq));
    }

    if (pq.empty()) {
        cout << "No characters provided.\n";
        return 0;
    }

    while (pq.size() > 1) {
        Node* left = pq.top(); pq.pop();
        Node* right = pq.top(); pq.pop();

        Node* parent = new Node('\0', left->freq + right->freq);
        parent->left = left;
        parent->right = right;
        pq.push(parent);
    }

    cout << "\nHuffman Codes:\n";
    printCodes(pq.top(), "");
    return 0;
}
```

Output

C++ Compiler — Output

```
Enter number of characters: 4
Enter character and frequency:
a 5
b 9
c 12
d 13

Huffman Codes:
d: 00
c: 01
a: 10
b: 11
```